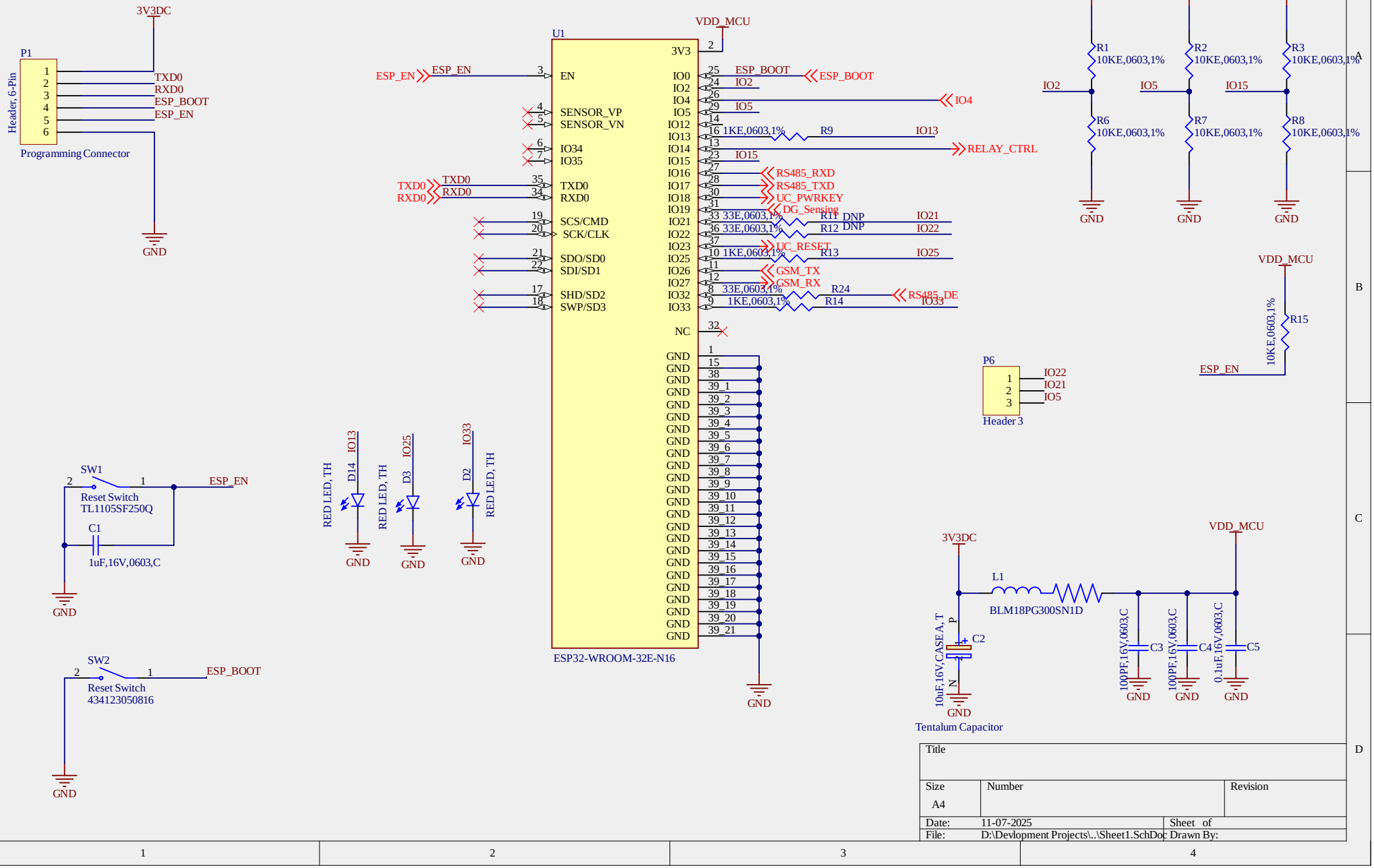
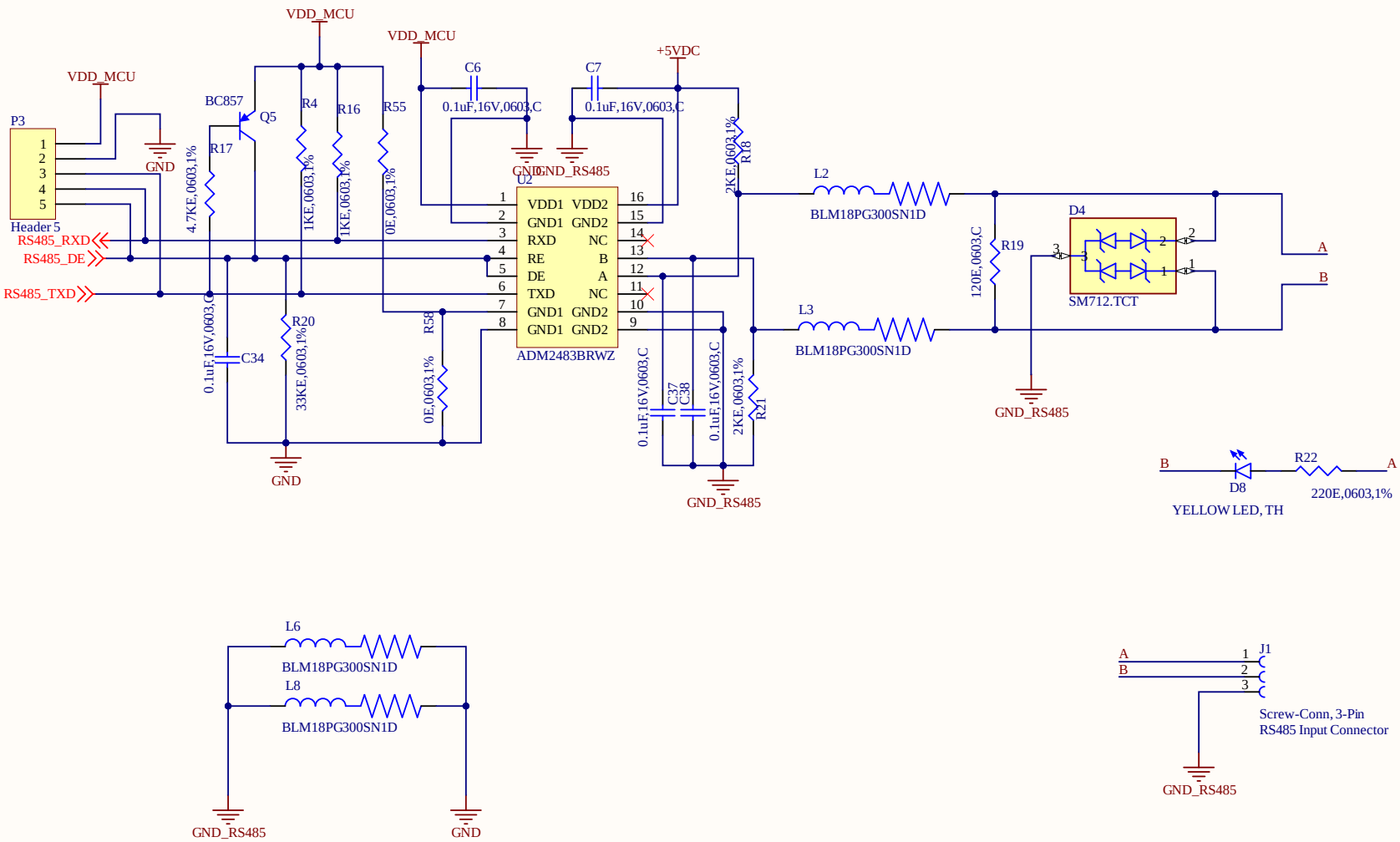
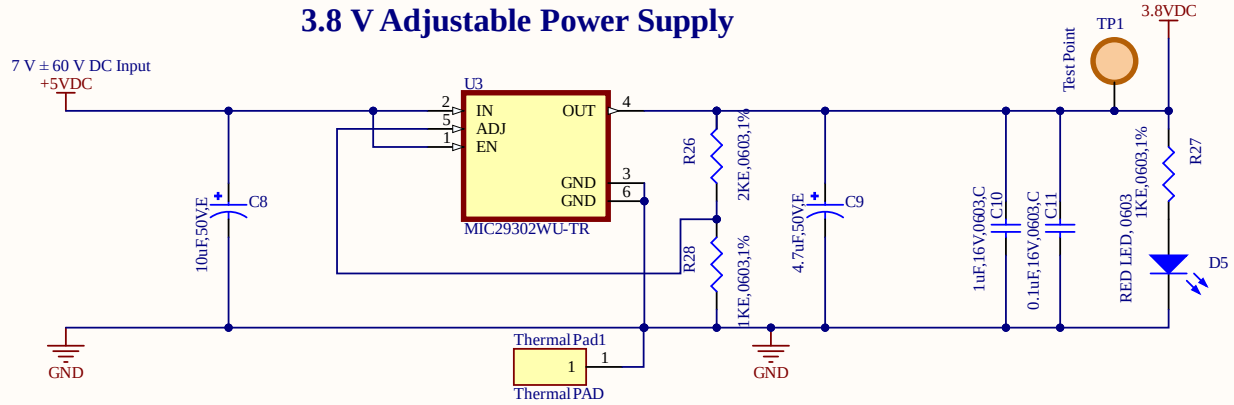
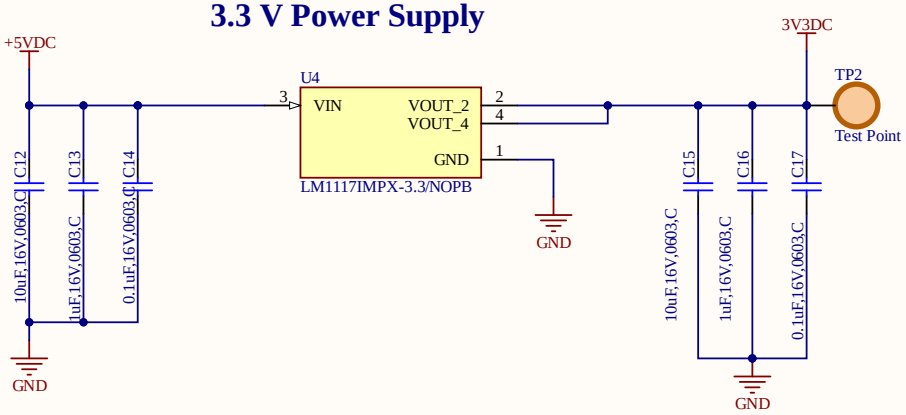

P1
Header, 6-Pin
Programming Connector
3V3DC
TXD0
RXD0
ESP_BOOT
ESP_EN
GND
U1
ESP32-WROOM-32E-N16
VDD_MCU
ESP_EN
TXD0
RXD0
SENSOR_VP
SENSOR_VN
IO34
IO35
SCS/CMD
SCK/CLK
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
NC
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
ESP_BOOT
RS485_RXD
RS485_TXD
UC_PWRKEY
DG_Sensing
UC_RESET
GSM_TX
GSM_RX
RS485_DE
RELAY_CTRL
1KE,0603,1%
R9
R10 DNP
R11 DNP
R12 DNP
R13
R14
R24
33E,0603,1%
R1
R2
R3
R6
R7
R8
10KE,0603,1%
R15
P6
Header 3
SW1
Reset Switch
TL1105SF250Q
C1
1uF,16V,0603,C
SW2
Reset Switch
434123050816
D14
D3
D2
RED LED, TH
L1
BLM18PG300SN1D
C2
C3
C4
C5
100PF,16V,0603,C
0.1uF,16V,0603,C
10uF,16V,CASE A, T
Tentalum Capacitor
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Size
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Number
Revision
Date:
11-07-2025
Sheet of
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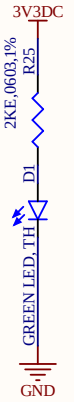

VDD_MCU
VDD_MCU
VDD_MCU
+5VDC
P3
Header 5
GND
BC857
Q5
R17
4.7KE,0603,1%
R4
1KE,0603,1%
R16
1KE,0603,1%
R55
0E,0603,1%
C6
0.1uF,16V,0603,C
C7
0.1uF,16V,0603,C
GND_RS485
U2
ADM2483BRWZ
VDD1 VDD2
GND1 GND2
RXD NC
RE B
DE A
TXD NC
GND1 GND2
GND1 GND2
RS485_RXD
RS485_DE
RS485_TXD
R18
2KE,0603,1%
L2
BLM18PG300SN1D
L3
BLM18PG300SN1D
R19
120E,0603,C
D4
SM712.TCT
A
B
GND_RS485
C34
0.1uF,16V,0603,C
R20
33KE,0603,1%
R58
0E,0603,1%
GND
C37
C38
0.1uF,16V,0603,C
0.1uF,16V,0603,C
2KE,0603,1%
R21
GND_RS485
B
D8
R22
A
220E,0603,1%
YELLOW LED, TH
L6
BLM18PG300SN1D
L8
BLM18PG300SN1D
GND_RS485
GND
A
B
J1
Screw-Conn, 3-Pin
RS485 Input Connector
GND_RS485
Title
Size
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Number
Revision
Date:
11-07-2025
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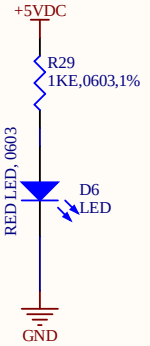
3.8 V Adjustable Power Supply

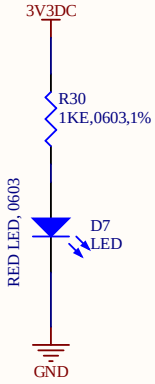


3.3 V Power Supply

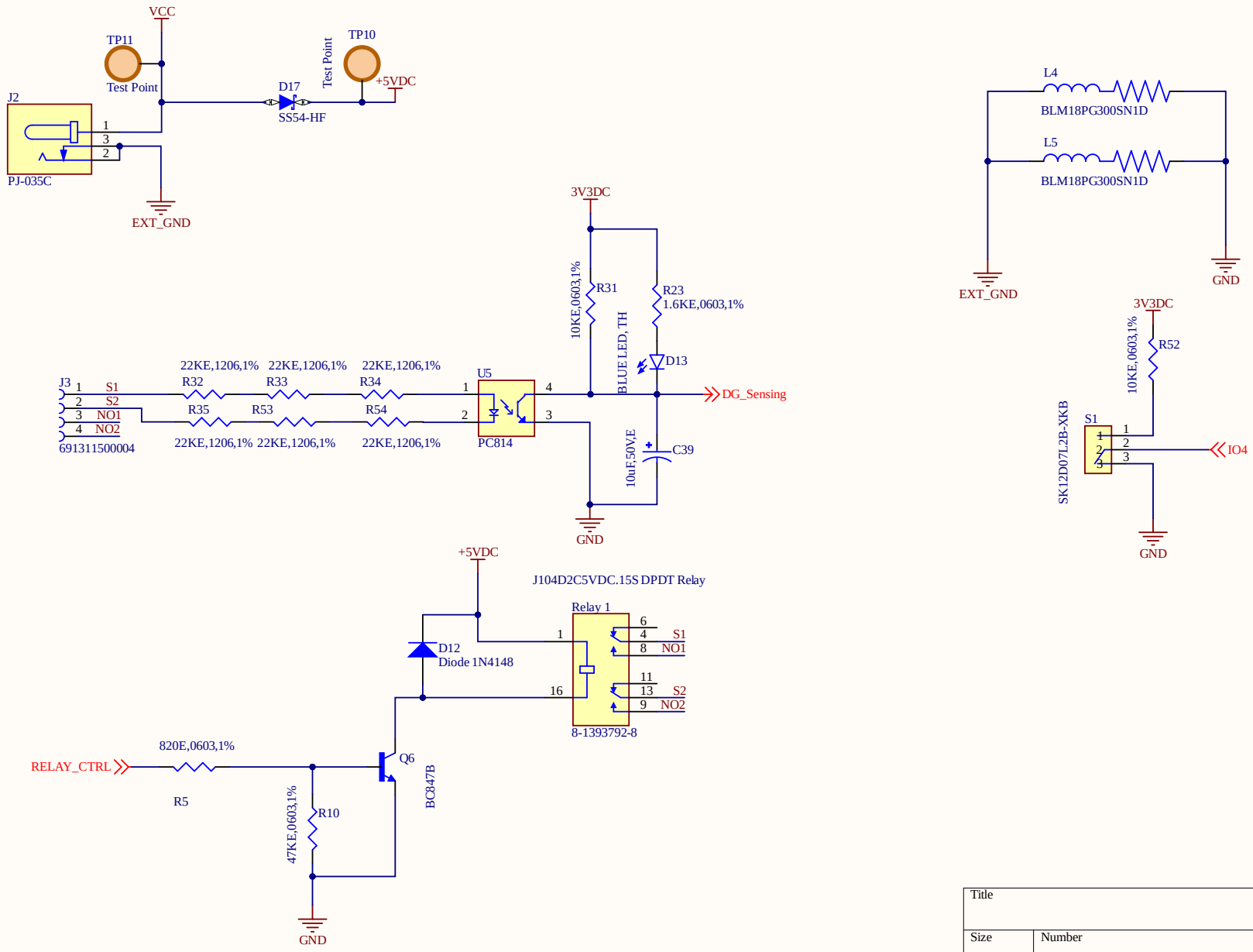




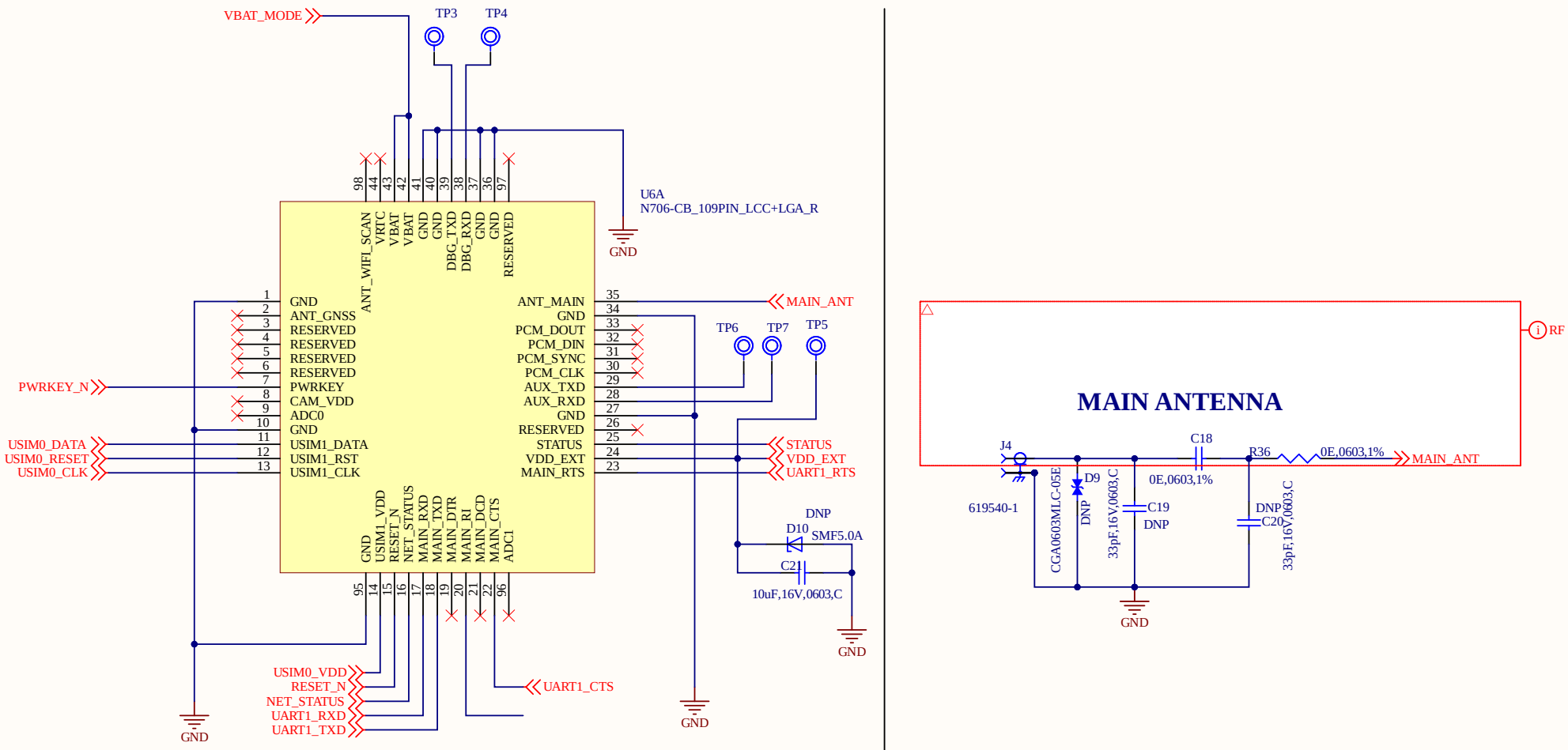




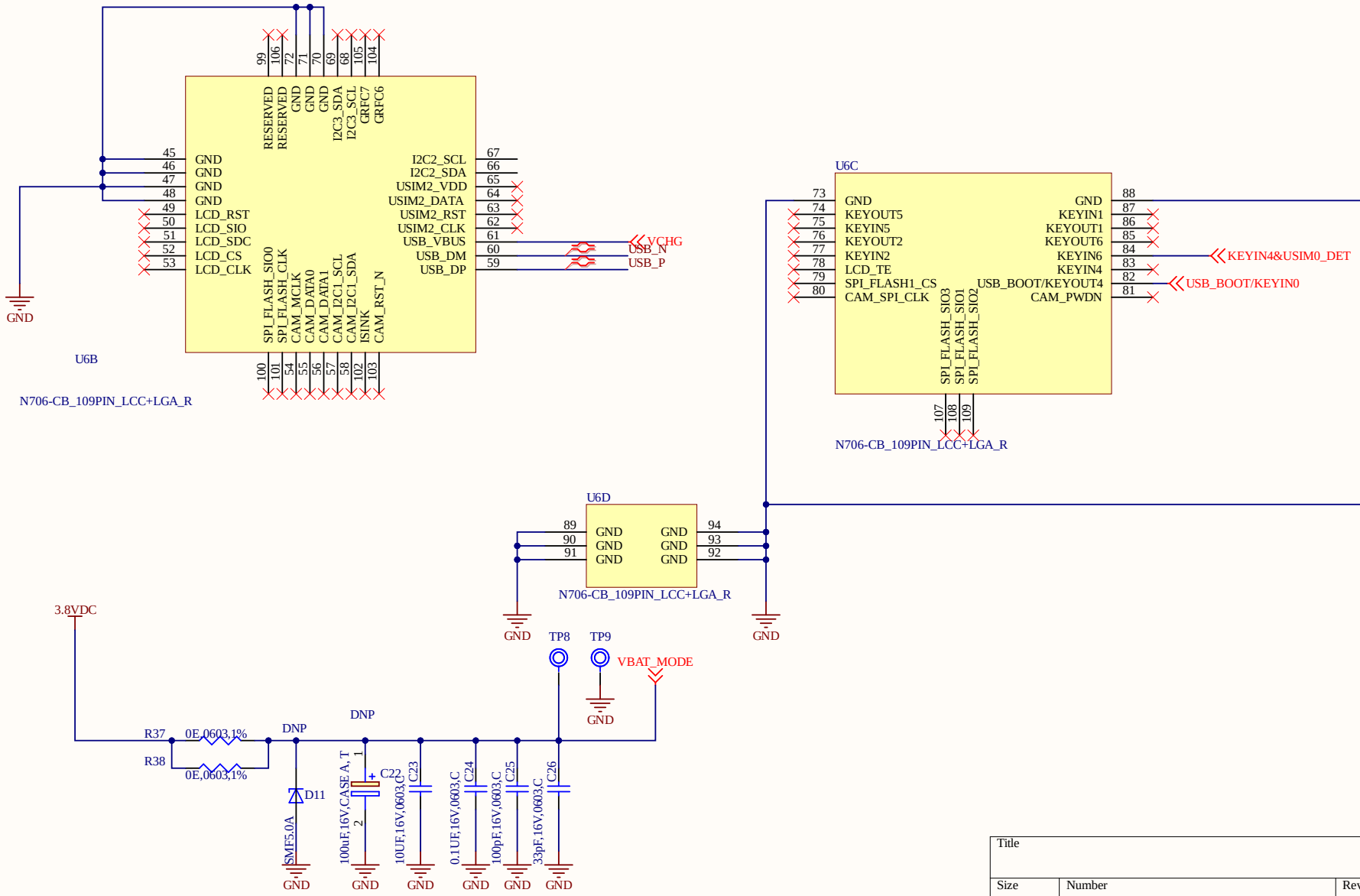
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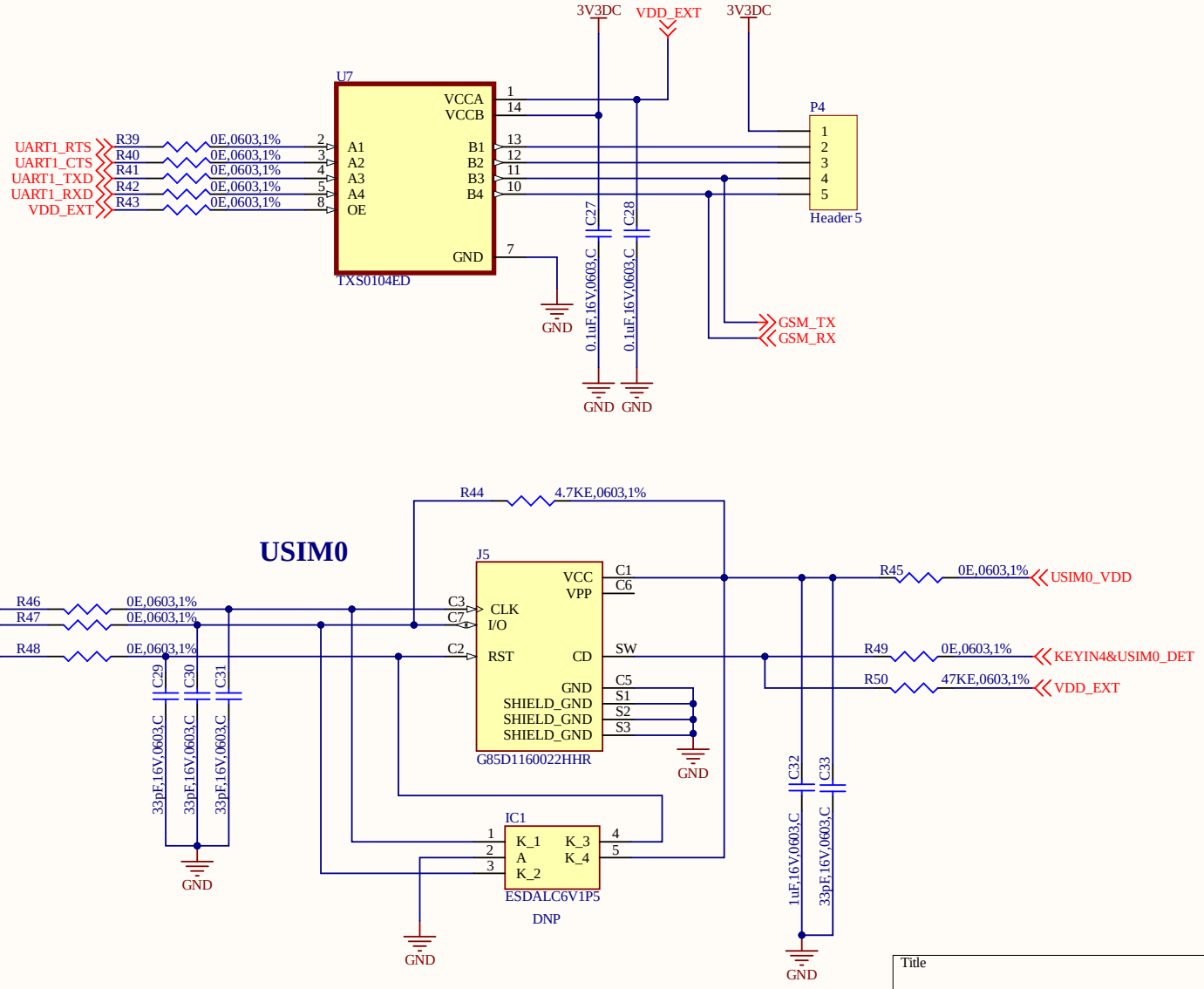


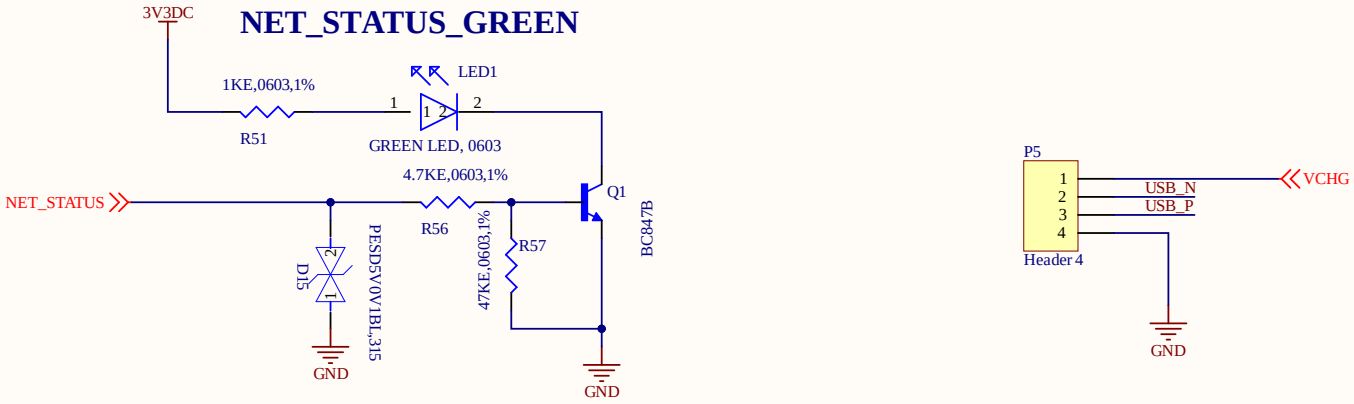
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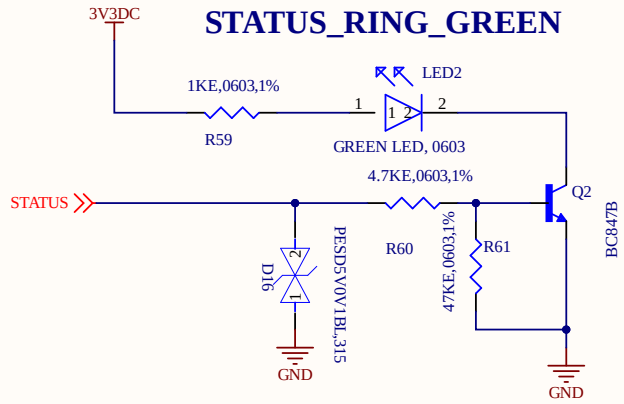


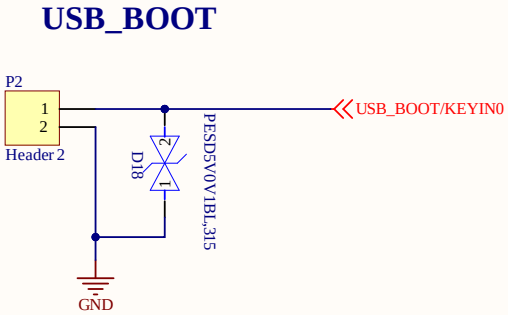
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U6B
N706-CB_109PIN_LCC+LGA_R
GND
RESERVED
RESERVED
GND
GND
GND
I2C3_SDA
I2C3_SCL
GRFC7
GRFC6
I2C2_SCL
I2C2_SDA
USIM2_VDD
USIM2_DATA
USIM2_RST
USIM2_CLK
USB_VBUS
USB_DM
USB_DP
LCD_RST
LCD_SIO
LCD_SDC
LCD_CS
LCD_CLK
SPI_FLASH_SIO0
SPI_FLASH_CLK
CAM_MCLK
CAM_DATA0
CAM_DATA1
CAM_I2C1_SCL
CAM_I2C1_SDA
ISINK
CAM_RST_N
VCHG
USB_N
USB_P
U6C
N706-CB_109PIN_LCC+LGA_R
KEYOUT5
KEYIN5
KEYOUT2
KEYIN2
LCD_TE
SPI_FLASH1_CS
CAM_SPI_CLK
KEYIN1
KEYOUT1
KEYOUT6
KEYIN6
KEYIN4
USB_BOOT/KEYOUT4
CAM_PWDN
SPI_FLASH_SIO3
SPI_FLASH_SIO1
SPI_FLASH_SIO2
KEYIN4&USIM0_DET
USB_BOOT/KEYIN0
U6D
N706-CB_109PIN_LCC+LGA_R
3.8VDC
TP8
TP9
VBAT_MODE
R37
R38
0E,0603,1%
DNP
D11
SMF5.0A
C22
100uF,16V,CASE A, T
C23
10UF,16V,0603,C
C24
0.1UF,16V,0603,C
C25
100pF,16V,0603,C
C26
33pF,16V,0603,C
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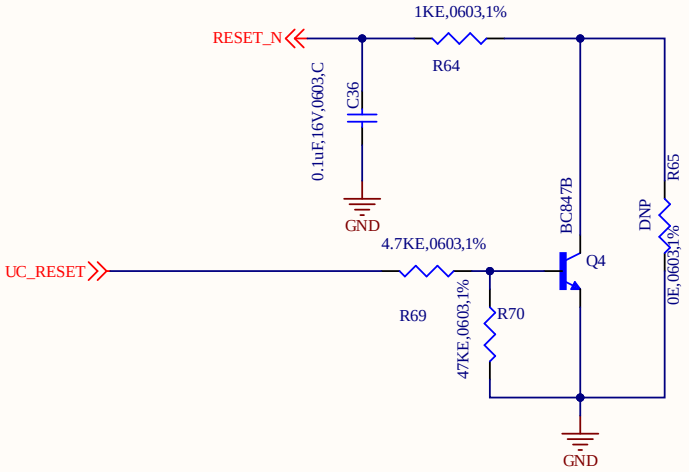

3V3DC
VDD_EXT
3V3DC
U7
VCCA
VCCB
A1
A2
A3
A4
OE
B1
B2
B3
B4
GND
TXS0104ED
UART1_RTS
UART1_CTS
UART1_TXD
UART1_RXD
VDD_EXT
R39
R40
R41
R42
R43
0E,0603,1%
P4
Header 5
C27
C28
0.1uF,16V,0603,C
GSM_TX
GSM_RX
USIM0
R44
4.7KE,0603,1%
J5
VCC
VPP
CLK
I/O
RST
CD
SW
GND
SHIELD_GND
G85D1160022HHR
USIM0_CLK
USIM0_DATA
USIM0_RESET
R46
R47
R48
C29
C30
C31
33pF,16V,0603,C
R45
USIM0_VDD
R49
KEYIN4&USIM0_DET
R50
47KE,0603,1%
VDD_EXT
IC1
K_1
K_2
K_3
K_4
A
ESDALC6V1P5
DNP
C32
C33
1uF,16V,0603,C
Title
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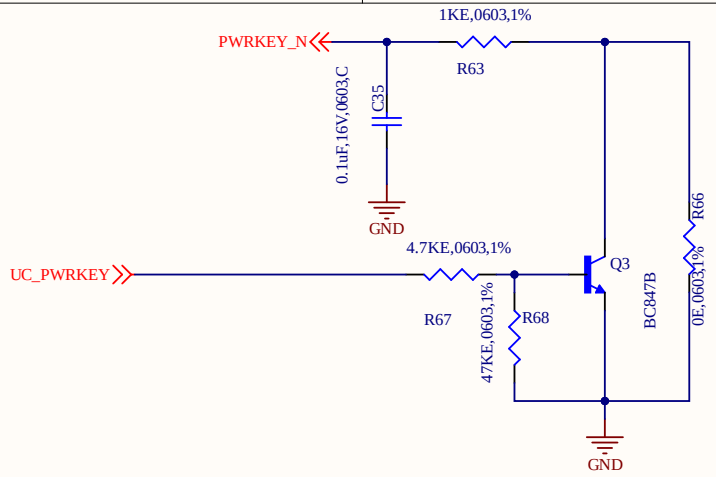




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RESET



PWRKEY

Title		
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File: D:\Devlopment Projects\..\Sheet9.SchDoc		Drawn By: